GlassQuill

Explainable Detection of AI-Generated Text

Case-Study Project Report (ESE)

Applications and Use Cases of Machine Learning (TE7483)
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Abstract

Large language models now write text that people cannot reliably tell apart from human writing. Colleges, newsrooms and hiring teams respond with AI-text detectors, but most of them return a bare score: a student who is wrongly flagged has no way to question it, and non-native English writers are flagged far more often than others. **GlassQuill** classifies an English text of 50 to 400 words as human-written or AI-generated and explains every decision.

We train four models that range from transparent to black box: a zero-shot GPT-2 perplexity threshold, TF-IDF with logistic regression, XGBoost on 18 interpretable style and perplexity features, and a fine-tuned DistilBERT. The data combines three public corpora, HC3, HAP-E and M4GT, covering ChatGPT (2022) and recent models such as GPT-4o, GPT-4, Llama-3, Mixtral and Gemma: 31,984 texts, balanced by class within each of 17 domains and split by question or document so that no topic leaks between training and testing. A first version scored above 99% because of formatting shortcuts in the data; removing them was the single most important step.

The deployed verdict, the mean of the two explainable models, reaches **97.4% accuracy, 0.974 F1 and a 2.0% false positive rate** on 4,829 unseen test texts, ahead of DistilBERT (96.1% accuracy, 7.1% false positive rate), and every verdict comes with SHAP feature reasons and word-level evidence. On MAGE, a benchmark of 27 generators never seen in training, every model falls to 61–66% accuracy, and paraphrasing lowers recall from 98% to 93%; we report both as limits. The project is implemented as a reproducible eight-stage pipeline and a tested web application.

Keywords: AI-generated text detection, explainable machine learning, perplexity, stylometry, XGBoost, SHAP, NLP text classification.

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1. Introduction and Problem Statement

1.1 Problem statement

Large language models (LLMs) such as ChatGPT, Gemini and Claude produce fluent text in seconds. Institutions that grade or publish writing have turned to automatic AI-text detectors, yet almost all of them output a single probability with no explanation. When such a detector is wrong, the cost falls on a real person: a student accused of cheating, or a writer whose work is rejected, cannot see or challenge the reasoning. Liang et al. (2023) showed that popular detectors flagged more than half of essays written by non-native English speakers as AI-generated [6].

Problem: build a detector that labels English text as human-written (0) or AI-generated (1), keeps the false positive rate (humans wrongly flagged as AI) low, and explains each decision in terms a non-expert can check.

1.2 Real-world relevance

- **Education:** universities use detectors in academic-integrity cases; an explanation turns a score into evidence that can be discussed.
- **Publishing and moderation:** platforms screen large volumes of submissions and need an audit trail for each decision.
- **Hiring:** cover letters and written tests are increasingly LLM-assisted, and employers need a fair second opinion.
- **Research:** most new detectors are black boxes; a transparent, well-documented baseline is useful for comparison.

1.3 Objective

Build and evaluate an explainable binary classifier for English text of 50 to 400 words that (a) achieves high accuracy with a low false positive rate, (b) explains every verdict through feature-level and word-level evidence, and (c) is tested for robustness on unseen generators and on paraphrased AI text; and deliver it as a reproducible pipeline and a working web application.

1.4 Research questions and hypotheses

ID	Research question
RQ1	Which simple signals (perplexity, sentence rhythm, vocabulary, punctuation) separate human and AI text, and how strongly?
RQ2	How close can an explainable model (XGBoost with SHAP) get to a black box (fine-tuned DistilBERT)?
RQ3	How well do the detectors work on unseen LLMs and on paraphrased AI text?
RQ4	Do two explanation methods (SHAP and LIME) agree on why a text was flagged?

Table 1: *Research questions*

Hypothesis	How it is tested
H1: AI text has lower perplexity	class means and SHAP direction of log_ppl
H2: AI text has lower burstiness (more even sentences)	class means and SHAP direction of burstiness and nll_std
H3: DistilBERT wins in-domain but drops most out-of-domain	F1 and FPR on the test set vs MAGE

Hypothesis	How it is tested
H4: paraphrasing lowers recall for every model	recall on AI texts before vs after T5 paraphrasing

Table 2: *Hypotheses*

1.5 Scope

- **In scope:** English text, 50 to 400 words, two classes (human, AI), many genres: question answering, essays, news, fiction, academic writing, blogs, peer reviews, Reddit, WikiHow and Wikipedia.
- **Out of scope:** other languages, source code, documents that mix human and AI writing, and identifying which LLM wrote a text.

1.6 Application domain and stakeholders

The application domain is **education and content moderation**, and the ML task is **binary NLP text classification**.

Stakeholder	What they need from GlassQuill
Teachers and universities	a second opinion that comes with reasons, never a final verdict
Students and writers	a fair way to see and question why their text was flagged
Content platforms and newsrooms	screening of large volumes of text with a clear audit trail
Researchers	a transparent, reproducible baseline to compare new detectors against

Table 3: *Stakeholders and their needs*

1.7 Refinement since Phase I

The Phase I proposal planned a study on HC3 alone, with XGBoost as the main model and a Gradio demo built in Google Colab. Building and testing it exposed two serious problems, and the project was refined as follows.

Area	Phase I plan	Final project	Why it changed
Data	HC3 only (ChatGPT, Dec 2022), about 12,000 texts	HC3 + HAP-E + M4GT: 31,984 texts, 17 domains, 2024 models	the HC3-only model labelled a modern ChatGPT paragraph 100% human
Preprocessing	basic cleaning	systematic shortcut removal (Table 8)	the first full run scored above 99% by spotting formatting, not style
Verdict	XGBoost alone	mean of XGBoost and TF-IDF + LR, chosen on validation	false positive rate fell from 6.6% to 2.0%
Model behaviour	unconstrained trees	monotone constraints where theory and data agree	prevents rules like 'very even sentences mean human'
Implementation	Colab notebook and Gradio demo	8-stage reproducible script, FastAPI + React web app, 28 automated tests	reproducibility, security and a usable interface

Table 4: *Refinement from the Phase I plan to the final project*

2. Background and Related Work

2.1 Key concepts

- **Perplexity:** the exponential of the mean negative log-probability of each token under a language model. LLMs choose likely tokens, so another language model finds their text easy to predict: AI text has lower perplexity. We use GPT-2 small [15] as the reader model.
- **Burstiness:** the standard deviation of sentence length divided by its mean. Human writers mix short and long sentences; LLM output is more even.
- **Stylometry:** measurable properties of writing style, such as vocabulary richness, word length, punctuation and function-word use.
- **TF-IDF with logistic regression:** each text becomes a weighted bag of words and word pairs; the linear model gives every n-gram an exact weight towards one class, so it is interpretable by construction.
- **XGBoost:** gradient-boosted decision trees [13], strong on tabular features and compatible with exact TreeSHAP explanations.
- **DistilBERT:** a compact transformer [14] fine-tuned end to end on the raw text; accurate but opaque.
- **SHAP:** Shapley values from cooperative game theory [10, 11]. The base value plus the sum of the SHAP values equals the model output, so every prediction is fully decomposed into feature contributions.
- **LIME:** perturbs the input, observes the prediction and fits a local linear model [12]; model-agnostic but approximate and random.

2.2 Related work

GLTR [7] visualised how often each token falls in a language model's top-k predictions, the idea behind our top-1 and top-10 features. DetectGPT [8] and Binoculars [9] detect AI text zero-shot from probability curvature and from the ratio of two models' perplexities. HC3 [1] introduced paired human and ChatGPT answers, MAGE [2] a 27-generator benchmark for detection in the wild, M4GT-Bench [4] multi-generator and multi-domain data, and HAP-E [3] parallel human and LLM continuations. RAID [5] showed that most detectors break under simple attacks such as paraphrasing. Few of these works explain individual decisions; GlassQuill combines a standard supervised setup with explanations, robustness tests and a strict focus on the false positive rate.

3. Dataset

3.1 Sources

Corpus	Source (Hugging Face)	Licence	AI writers	Domains	Split group
HC3	Hello-SimpleAI/HC3	CC BY-SA 4.0	ChatGPT (2022)	Reddit ELI5, finance, medicine, open QA, CS Wikipedia	question
HAP-E	browndw/human-ai-parallel-corpus	MIT	GPT-4o, GPT-4o-mini, Llama-3 8B/70B	academic, news, fiction, spoken, blogs, TV/movie scripts	document
M4GT	Jinyan1/COLING_2025_MGT_en (M4GT part)	per source	GPT-4o, GPT-4, GPT-3.5, Mixtral, Llama-3, Gemma, Cohere	arXiv, essays, peer reviews, Reddit, WikiHow, Wikipedia	text
MAGE (test only)	yaful/MAGE (test split)	Apache 2.0	27 models incl. OPT, BLOOM, GPT-J, Flan-T5, LLaMA, GPT-3/3.5/4	14 domains	not used in training

Table 5: *Dataset sources*

The HC3 and MAGE parts of the COLING data are excluded so that MAGE remains a clean out-of-domain test. All corpora are downloaded automatically by the pipeline.

3.2 Attributes, target variable and data types

Attribute	Data type	Role	Description
text	string	input	the passage to classify
label	integer (0/1)	target	0 = human-written, 1 = AI-generated
qid	string	grouping	question or source-document id; used only to split the data
source	categorical (17)	analysis	domain, e.g. hc3 finance, hape/news, m4gt/arxiv
generator	categorical	analysis	'human' or the name of the model that wrote the text
corpus	categorical (3)	analysis	HC3, HAP-E or M4GT
n_words	integer	filter	number of words after cleaning (50 to 400)

Table 6: *Attributes, target variable and data types*

The models receive either the raw text (TF-IDF + LR, DistilBERT) or 18 numeric features engineered from it (XGBoost, section 3.6). The group id is never a model input.

3.3 Size, balance and exploratory analysis

Split	Texts	HC3	HAP-E	M4GT	Share AI
Train (70%)	22,371	8,412	5,565	8,394	0.500
Validation (15%)	4,784	1,776	1,209	1,799	0.500
Test (15%)	4,829	1,808	1,222	1,799	0.501
Total	**31,984**	**11,996**	**7,996**	**11,992**	**0.500**
MAGE (out-of-domain)	2,000	–	–	–	0.500

Table 7: *Dataset size per split and corpus*

Every one of the 17 domains contains exactly as many human as AI texts (Appendix B). Mean length is 256 words for human and 266 words for AI texts, so length alone is not a shortcut (Figure 1); the peak at 400 words comes from long texts being cut at the last full sentence.

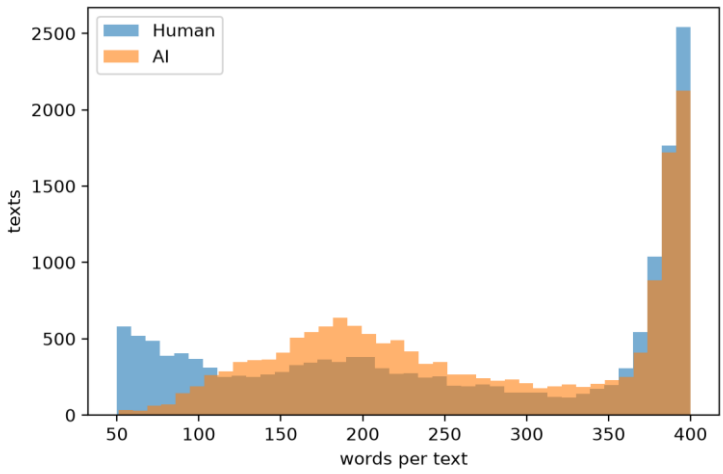


Figure 1: *Distribution of text length (words) for human and AI texts*

3.4 Missing values and duplicates

- Rows with empty or missing text are dropped (dropna), and exact duplicate texts are removed.
- After cleaning there are **0 missing values** in any split; every attribute is fully populated.
- The engineered features cannot be missing: every rate divides by max(words, 1), and texts are at least 50 words long.

3.5 Preprocessing and shortcut removal

The cleaning function (clean_text in src/data.py) removes formatting that reveals the source of a text rather than its writing style. Each artifact below appeared mostly in one class and would let a model cheat; together they explained why the first full run scored 99.3%.

Artifact	Human texts	AI texts	Fix
literal "\n" line breaks (HC3 ChatGPT)	0%	most	replace with a space
"do n't", "it 's" (tokenizer split)	34%	0.02%	rejoin
"(word)" spaced brackets	35%	0.01%	remove the spaces
URL_0 link placeholders	11.6%	0.07%	remove
markdown bold and # headings (M4GT)	2%	9%	remove
curly quotes, em dashes, [1] citations	more often	less often	normalise to plain ASCII
"As an AI language model…", "I hope this helps"	0%	some	remove

Table 8: *Shortcut artifacts removed during preprocessing*

Further steps: texts outside 50–400 words are removed, while HAP-E and M4GT texts longer than 400 words are cut at the last full sentence so that long texts of both classes are kept the same way. Each domain is then **balanced** to equal human and AI counts; without this, HC3 open_qa had 20 human vs 764 AI texts and the topic alone predicted the label. Finally, MAGE's label convention (1 = human) is flipped to ours (1 = AI).

3.6 Feature engineering

XGBoost uses 18 hand-crafted features. Six come from GPT-2 (how predictable each token is) and twelve describe writing style. Table 9 lists them with their class means on the training set and the effect size (difference of class means in pooled standard deviations).

Group	Feature	Meaning	Human	AI	Effect
GPT-2	log_ppl	log perplexity (mean surprise)	3.57	2.61	−1.39
	top1_frac	share of tokens that were GPT-2's first guess	0.34	0.46	+1.36
	top10_frac	share of tokens in GPT-2's top 10	0.67	0.79	+1.35
	nll_std	unevenness of surprise	2.90	2.37	−1.25
	mean_entropy	GPT-2's uncertainty	3.71	3.27	−1.00
	nll_max	most surprising token	13.8	11.7	−0.92
Rhythm	burstiness	std / mean sentence length	0.53	0.38	−0.72
	std_sent_len	sentence length spread	11.0	8.1	−0.40
	avg_sent_len	mean sentence length	21.6	22.5	+0.07
Vocabulary	type_token_ratio	unique / total words	0.59	0.52	−0.62
	function_word_rate	the, of, and, …	0.28	0.30	+0.50
	avg_word_len	letters per word	4.64	4.87	+0.42
Punctuation	punct_rate	punctuation per word	0.19	0.16	−0.37
	comma_rate	commas per word	0.050	0.058	+0.27
Voice	contraction_rate	don't, it's, …	0.018	0.014	−0.19
	first_person_rate	I, we, my, …	0.016	0.012	−0.16
Readability	flesch_reading_ease	Flesch reading ease	56.6	49.0	−0.33
	fk_grade	Flesch-Kincaid grade	10.3	11.5	+0.25

Table 9: *The 18 engineered features, class means and effect sizes (training set)*

3.7 Feature selection

All 18 features are kept. Each was chosen in advance with a stated hypothesis about how AI text differs, tree ensembles are robust to correlated features, and SHAP (section 6.6) shows afterwards which features the model actually relies on. For ten features where theory and the training data agree on the direction (for example, lower perplexity and lower burstiness indicate AI), XGBoost is given **monotone constraints**, which acts as a soft form of feature control: these features may only push the prediction in the expected direction.

3.8 Validation strategy

- **Group split 70/15/15** (GroupShuffleSplit) inside each corpus: all answers to one HC3 question, or all versions of one HAP-E document, land in the same split. The absence of overlap is asserted in code.
- **5-fold GroupKFold** on the training set only, for hyper-parameter tuning, so tuning is leak-free too.
- **Validation set** for the perplexity threshold, the DistilBERT epoch and the choice of the deployed verdict rule.
- **Test set** used once, at the end. **MAGE** and the **paraphrase set** measure robustness.
- **Fixed seed 42** for Python, NumPy, PyTorch, the splits and all sampling.

4. Methodology and ML Pipeline

4.1 Pipeline overview

Figure 2 shows the complete pipeline. Each numbered step is a function in run_pipeline.py; every step saves its output, so later steps can be rerun on their own.

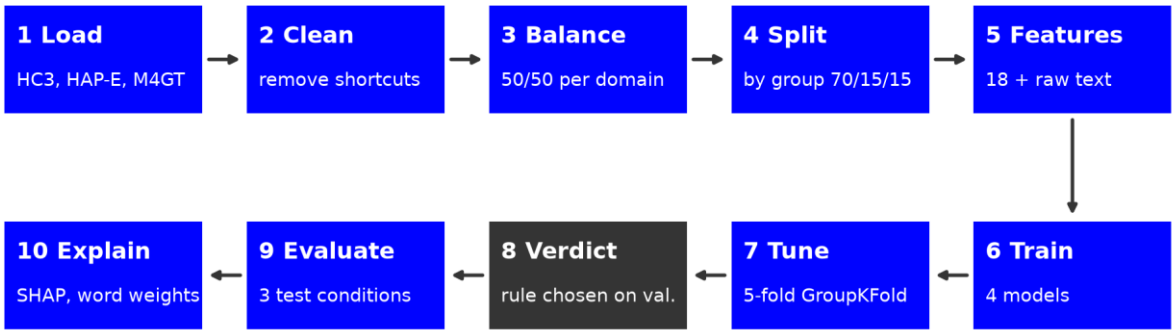


Figure 2: *The GlassQuill ML pipeline (read the top row left to right, then the bottom row right to left)*

4.2 Model selection and justification

The four models deliberately range from simple and transparent to powerful and opaque. This lets us measure what explainability costs (RQ2) instead of assuming it.

#	Model	Input	Why it is included	Explained with
1	Perplexity threshold	log_ppl	zero-shot baseline: how far a single, well-known signal goes	the value itself
2	TF-IDF (1–2-grams, 50k) + logistic regression	words	classic glass-box text model, strong on wording	exact weights (coefficient × TF-IDF)
3	XGBoost (400 trees)	18 features	strong on tabular data; every feature has a meaning	TreeSHAP (exact)
4	DistilBERT (fine-tuned)	raw text, 256 tokens	strong black box to compare against	LIME, SHAP text
5	Deployed verdict = mean of 2 and 3	both	the two glass boxes make different mistakes, so averaging cancels errors	both explanations

Table 10: *Models and the reason each one is included*

4.3 Training strategy and hyper-parameters

Model	Training and tuning	Selected
Perplexity threshold	cut-off on validation by Youden's J (maximise TPR − FPR)	AI if log_ppl ≤ 2.98
TF-IDF + LR	C ∈ {0.1, 1, 10}; 5-fold GroupKFold; scored on F1	C = 10
XGBoost	max_depth ∈ {3, 4, 6}, learning rate ∈ {0.05, 0.1}, 400 trees, subsample and colsample 0.8, monotone constraints; same CV	best grid point by F1
DistilBERT	learning rate 2e-5, batch 16, 2 epochs, max 256 tokens, weight decay 0.01	epoch with best validation F1 (0.964)
Verdict rule	compared XGBoost, LR and their mean on the validation set	mean: accuracy 0.972, FPR 0.021

Table 11: *Training strategy and selected hyper-parameters*

On the validation set the mean of XGBoost and LR reached accuracy 0.972 at FPR 0.021, against 0.929 / 0.064 for XGBoost alone and 0.954 / 0.032 for LR alone, so it was chosen as the deployed rule before the test set was touched. The app reports AI for $p \geq 0.65$, human for $p \leq 0.35$, and 'unclear' in between, so weak evidence is shown as such instead of being forced into a class.

4.4 Evaluation method

Every model is scored by the same function on the same texts with **accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, ROC curves and the false positive rate (FPR)**. FPR is reported next to every other metric because a false positive accuses a real person; accuracy alone can hide very different error profiles. Three test conditions are used: (a) the in-domain test set, overall and per corpus, (b) MAGE, per generator and per domain, and (c) AI texts before and after paraphrasing every sentence with a T5 paraphraser (humarin/chatgpt_paraphraser_on_T5_base).

4.5 Explanation method

- **Global:** TreeSHAP over the whole test set (beeswarm and bar plots) and the direction in which each feature pushes the prediction.
- **Local:** waterfall plots for one correct AI text, one correct human text, one false positive and one false negative.
- **Words:** exact per-word contributions of the logistic regression, shown as coloured text in the app.
- **Agreement checks (RQ4):** SHAP vs LIME on DistilBERT, and LIME vs the exactly known LR weights.

5. Implementation

5.1 Tools and environment

Python 3.11, PyTorch 2.11 (CUDA 12.8), Hugging Face Transformers 5.20, scikit-learn 1.9, XGBoost 3.2, SHAP 0.51, LIME 0.2, textstat and pandas; FastAPI for the API; React 19, TypeScript, Tailwind CSS and Motion for the website; pytest, ESLint and the TypeScript compiler for quality checks. Hardware: a laptop with an NVIDIA RTX 3050 (4 GB).

5.2 Project structure

Path	Responsibility
src/data.py	dataset URLs, loaders, clean_text, balancing (prepare), group_split
src/features.py	12 stylometric features and the GPT-2 PerplexityScorer (6 features)
src/models.py	metrics (evaluate), perplexity threshold, TF-IDF + LR and XGBoost training with GroupKFold
src/train_distilbert.py	DistilBERT fine-tuning and batched prediction
src/explain.py	SHAP plots, exact word weights, highlights, LIME
src/robustness.py	T5 paraphraser for the paraphrase attack
src/detector.py	loads the saved models once and analyses one text for the app
run_pipeline.py	the whole study: 8 stages, cached outputs, --quick mode, fixed seed
server.py	FastAPI app: input validation, rate limiting, security headers, serves the website
web/	React website: home, detector, results and method pages
tests/test_server.py	28 automated API tests
models/, reports/	trained models; result tables, figures and run logs

Table 12: *Project structure*

5.3 Workflow and reproducibility

The full study runs with one command. Each stage writes its outputs to disk and is skipped or reused on the next run, so any stage can be repeated alone:

```
python run_pipeline.py --quick      # all 8 stages on a small sample, about 10 minutes
python run_pipeline.py              # the full study, about 1.5 hours on the laptop GPU
python run_pipeline.py --stages evaluate ood paraphrase explain
python -m pytest tests -q           # 28 API tests
python server.py                    # web app at http://127.0.0.1:8000
```

The stages are data → features → train → bert → evaluate → ood → paraphrase → explain. Datasets download automatically, GPT-2 features are cached per split, and all result tables are collected into reports/RESULTS.md. The trained models are committed, so the web app runs without retraining.

5.4 The web application

A user pastes a text; the server removes invisible characters, checks that the input is English prose of at least 50 words, computes the 18 features, runs XGBoost and the word model, and returns the verdict with an 'unclear' band, both model scores, the top reasons compared with typical human and AI values, a SHAP chart and word-level evidence (Figures 3 and 4). Each analysis takes about 0.1 s on the GPU. Pasted text is never stored or logged.

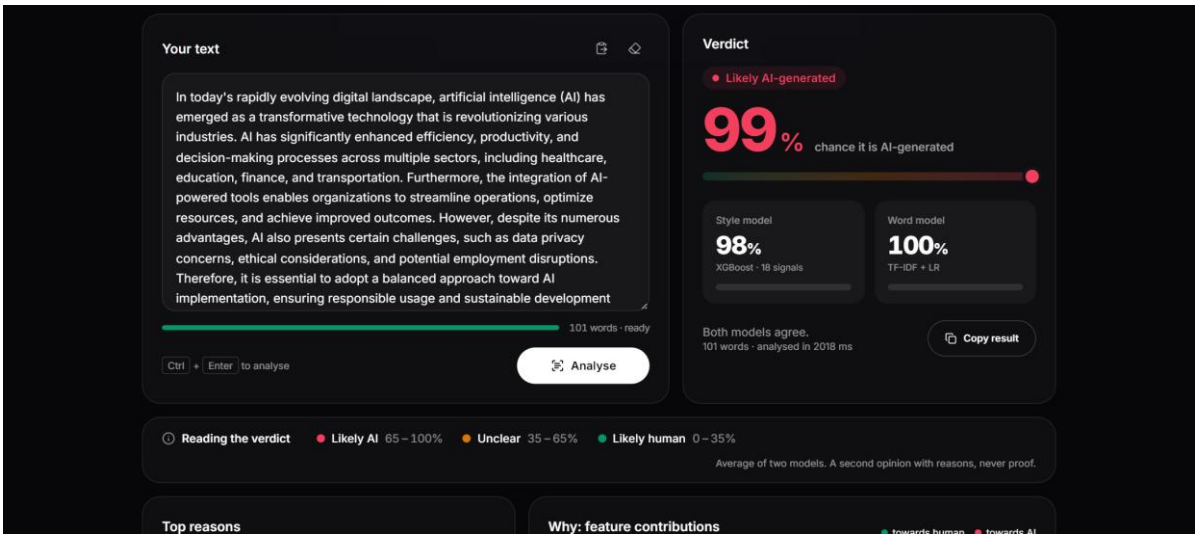


Figure 3: *Detector page: verdict, unclear band and the scores of both models*

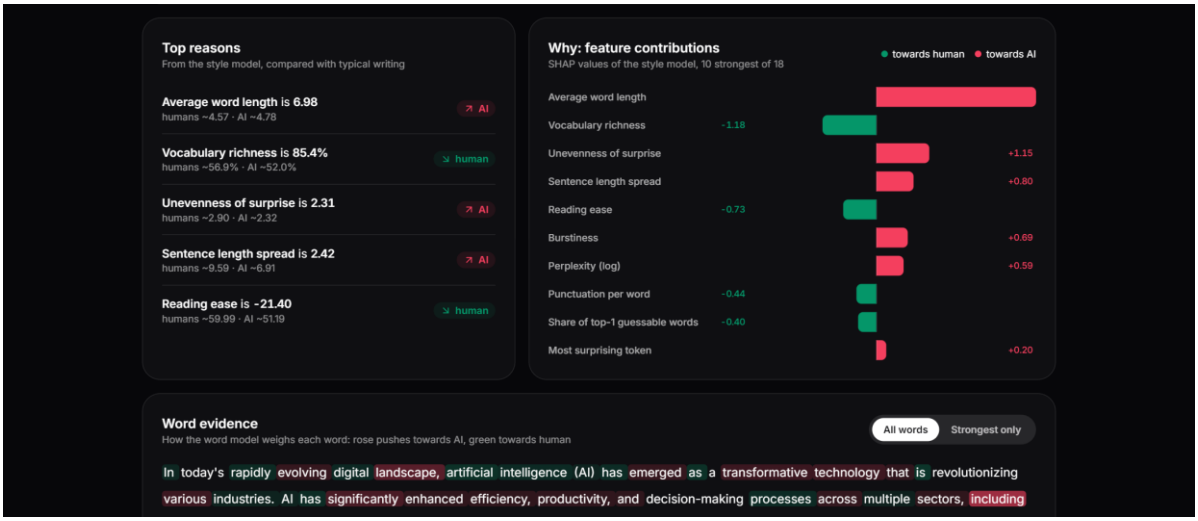


Figure 4: *Explanations in the app: top reasons and SHAP feature contributions*

5.5 Testing and security

- 28 automated tests cover input validation (too short, too long, wrong type, not English), behaviour (a known AI text is flagged and a human text is not; invisible characters do not change the verdict) and security.
- Security measures: request size cap, mandatory Content-Length, per-IP rate limit (20 analyses per minute), a strict Content-Security-Policy and other security headers, path-traversal protection, and error messages that never echo the submitted text.

6. Results and Comparative Analysis

6.1 In-domain test set

Model	Accuracy	Precision	Recall	F1	ROC-AUC	FPR
Perplexity threshold	0.839	0.892	0.771	0.827	0.902	0.093
TF-IDF + LR	0.959	0.970	0.948	0.959	0.990	0.030
XGBoost	0.923	0.933	0.912	0.923	0.978	0.066
DistilBERT	0.961	0.934	0.993	0.963	0.997	0.071
GlassQuill verdict (XGBoost + LR)	**0.974**	**0.980**	**0.968**	**0.974**	**0.995**	**0.020**

Table 13: *Results on the in-domain test set (4,829 texts)*

The deployed verdict has the best accuracy, precision and F1 and the lowest false positive rate. DistilBERT has the highest recall and ROC-AUC, but it wrongly flags 7.1% of human texts, three and a half times as many as the verdict. Averaging the two explainable models improves on both of them: XGBoost and the word model make different mistakes.



Figure 5: *Confusion matrices on the test set (rows: true class, columns: predicted class)*

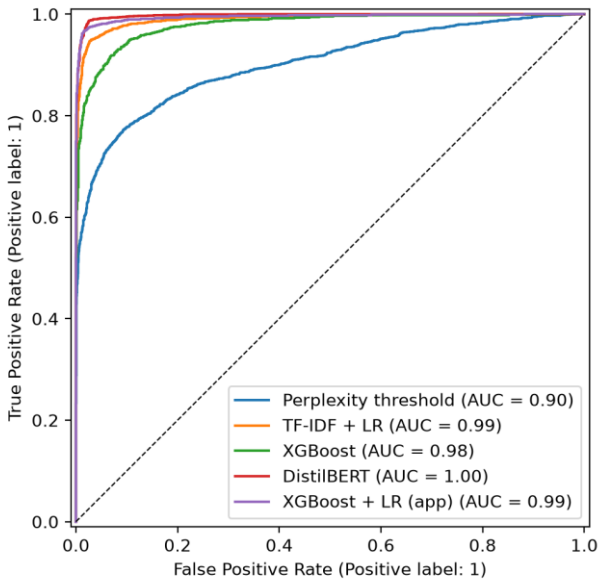


Figure 6: *ROC curves on the test set*

6.2 Results per corpus

Model	HC3	HAP-E	M4GT
Perplexity threshold	0.958	0.606	0.800
TF-IDF + LR	0.973	0.977	0.932
XGBoost	0.964	0.893	0.898

Model	HC3	HAP-E	M4GT
DistilBERT	0.970	0.973	0.949
GlassQuill verdict	**0.987**	**0.981**	**0.956**

Table 14: *F1 score per corpus on the test set*

The perplexity threshold works well on 2022 ChatGPT text (HC3) but drops to 0.61 F1 on HAP-E, where LLMs continue a human text in the same style. The verdict is the best model on every corpus.

6.3 Out-of-domain: MAGE

Model	Accuracy	Precision	Recall	F1	ROC-AUC	FPR
Perplexity threshold	0.656	0.792	0.422	0.551	0.681	0.111
TF-IDF + LR	0.606	0.787	0.292	0.426	0.648	0.079
XGBoost	0.634	0.736	0.416	0.532	0.611	0.149
DistilBERT	0.655	0.734	0.486	0.585	0.696	0.176
GlassQuill verdict	**0.664**	**0.854**	**0.397**	**0.542**	**0.655**	**0.068**

Table 15: *Results on MAGE: 2,000 texts from 27 generators and 14 domains never used in training*

Every model falls to 61–66% accuracy on unseen generators, which confirms that AI-text detection does not transfer easily. The verdict still has the best accuracy and precision and by far the lowest false positive rate (6.8%), while DistilBERT's FPR rises to 17.6%. Accuracy per generator family shows where the errors come from: **GPT-4 0.91**, GLM-130B 0.81, GPT-3.5 0.80, GPT-3 davinci 0.76, LLaMA 0.54, GPT-4 paraphrase 0.49, T0 0.47, Flan-T5 0.35, OPT 0.04, BLOOM 0.02, GPT-J/NeoX 0.00, and human texts 0.93. Instruction-tuned chat models, the ones students actually use, are caught; small base models that sample unusual words look human to every detector. Compared with the HC3-only Phase I model, GPT-4 detection rose from 0.55 to 0.91 and GPT-3.5 from 0.55 to 0.80.

6.4 Paraphrase attack

Model	Recall before	Recall after	Drop
Perplexity threshold	1.00	0.93	−0.07
TF-IDF + LR	0.93	0.77	−0.17
XGBoost	1.00	0.95	−0.05
DistilBERT	1.00	1.00	0.00
GlassQuill verdict	**0.98**	**0.93**	**−0.05**

Table 16: *Recall on 60 AI test texts before and after paraphrasing every sentence*

Paraphrasing hurts the word model most (−17 points) because the exact words change; the perplexity and style features mostly survive (−5 to −7 points). DistilBERT is unaffected on this small set. The verdict still catches 93% of paraphrased AI texts.

6.5 Comparative analysis

- **Explainable vs black box (RQ2):** XGBoost alone trails DistilBERT (0.923 vs 0.963 F1), but the explainable verdict beats it (0.974 F1) with less than a third of its false positives, on every corpus.

- **Precision–recall trade-off:** DistilBERT is recall-oriented (0.993) at the cost of precision (0.934); the verdict is balanced (0.980 / 0.968), which is the right profile when false accusations are the costly error.
- **Robustness:** in-domain differences shrink out-of-domain, where all models land between 61% and 66%; the verdict's advantage there is its low FPR. Paraphrasing separates the models by what they read: words are fragile, style and perplexity are robust.
- **Data quality vs model choice:** removing shortcuts changed results more than any model change: the uncleaned data gave a misleading 99.3%, and adding modern corpora fixed the failure on current ChatGPT output.

6.6 Explanations

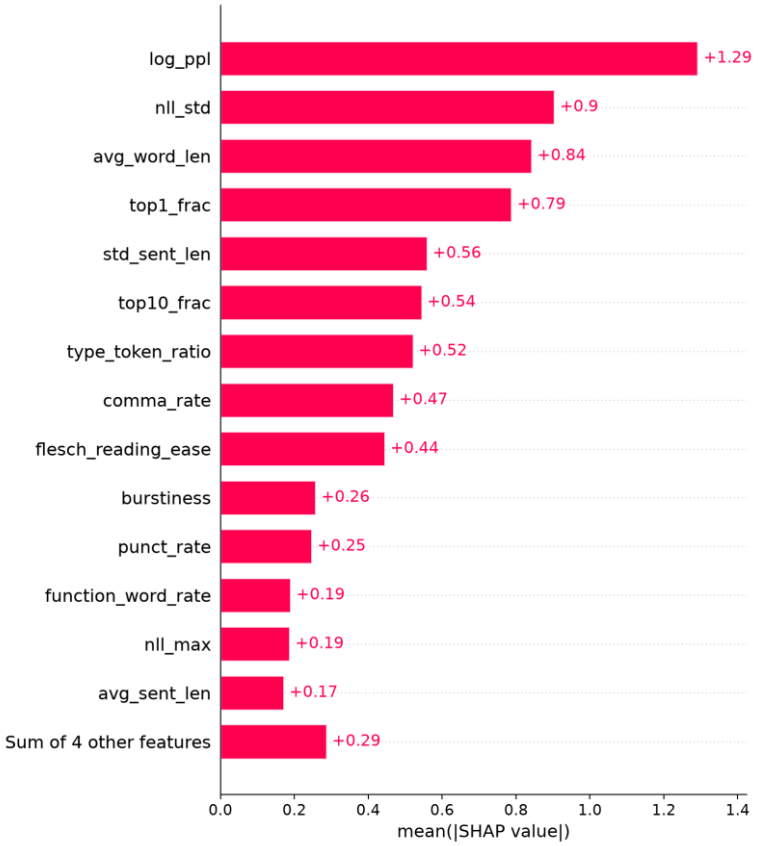


Figure 7: *Global feature importance of XGBoost (mean |SHAP| over the test set)*

- **Global SHAP:** the most important features are log perplexity (low → AI), nll_std (even surprise → AI), average word length (long words → AI), top1_frac (guessable words → AI) and sentence-length spread (even → AI).
- **Glass-box words:** the strongest AI words of the logistic regression include 'including', 'important to', 'overall' and 'significant'; human words include 'etc', 'basically', 'since' and 'thus'.
- **SHAP vs LIME on DistilBERT:** top-10 word overlap (Jaccard) is only 0.06, partly because DistilBERT works on word pieces while LIME removes whole words.
- **LIME vs exact LR weights:** LIME recovers 51% of the true top-10 words, so the app uses exact SHAP values and exact word weights instead.

6.7 Answers to the research questions

RQ	Answer
RQ1	GPT-2 surprise features separate the classes best (about 1.3 SD: log perplexity, top-1/top-10 share, nll_std). Burstiness and vocabulary richness help moderately; sentence length alone does not. H1 and H2 are confirmed by both class means and SHAP direction.
RQ2	An explainable system can match the black box: the verdict reaches 0.974 F1 at 2.0% FPR vs DistilBERT's 0.963 F1 at 7.1% FPR.
RQ3	All models drop to 61–66% accuracy on unseen generators, but chat models are still caught (GPT-4 91%). Paraphrasing lowers recall for every model except DistilBERT (H4 mostly holds). H3 is partly confirmed: DistilBERT is not the best in-domain on accuracy, but has the highest FPR on MAGE (17.6%).
RQ4	No. SHAP and LIME barely agree on DistilBERT (Jaccard 0.06), and LIME recovers only half of the known LR words.

Table 17: *Answers to the research questions*

7. Limitations and Ethical Considerations

- A detector score must **never** be the only evidence against a person. The app says so on every page and shows an 'unclear' band instead of guessing.
- Simple, predictable wording, common in non-native writing, has low perplexity and risks false positives. FPR is reported everywhere, but fairness on non-native essays was not tested directly.
- Small open models with sampled output (OPT, BLOOM, GPT-J) and paraphrasing tools fool all four models. New LLMs appear quickly and can make results stale.
- English only, and only the genres in the training corpora. Texts that mix human and AI writing are not handled.
- HAP-E's AI texts continue a human text in the same style, which is harder than detecting a normal chatbot answer, so results on it are a lower bound.
- Privacy: submitted text is processed in memory and never stored or logged.

8. Conclusion and Future Scope

8.1 Conclusion

GlassQuill shows that an explainable detector can match a fine-tuned transformer. Averaging an XGBoost model on 18 meaningful features with a word-level logistic regression gives 97.4% accuracy, 0.974 F1 and a 2.0% false positive rate on 4,829 unseen texts, ahead of DistilBERT, and every decision comes with feature-level and word-level reasons. Two findings mattered most: removing formatting shortcuts from the data turned a misleading 99.3% into an honest result, and training on modern LLM output with direction-constrained features fixed the failure of the Phase I model on current ChatGPT text. The main weaknesses are unseen generator families and paraphrasing. The project is delivered as a reproducible pipeline, a tested web application, and complete documentation.

8.2 Future scope

1. Add a stronger reader model or Binoculars (a two-model perplexity ratio) as extra features; an early test gave +0.01 to +0.04 AUC at about 25 times the cost.
2. Set the decision threshold for a fixed, very low false positive rate (for example 1%) instead of 0.5.
3. Sentence-level scoring to locate AI-written passages inside human texts.
4. Evaluate on the RAID benchmark and add newer generators to training.
5. Calibrate probabilities so that '80%' means 80%, and test fairness on non-native English essays.
6. Extend to Indian languages.

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Appendix A. Rubric Coverage

ESE component (marks)	Where it is addressed
Problem refinement and use case relevance (3)	Ch. 1: problem, relevance, objective, scope, domain, stakeholders, refinement since Phase I (Table 4)
Dataset understanding, preprocessing, feature handling (4)	Ch. 3: sources, size, attributes, target, data types, missing values, preprocessing, feature engineering and selection, validation strategy
Methodology and ML pipeline design (5)	Ch. 4: pipeline figure, model selection and justification, training strategy, evaluation and explanation methods
Implementation / project execution (5)	Ch. 5: tools, project structure, reproducible workflow, web application, tests and security
Results, evaluation metrics, comparative analysis (5)	Ch. 6: accuracy, precision, recall, F1, ROC-AUC, FPR, confusion matrices, ROC curves, per-corpus, out-of-domain, paraphrase, explanations, RQ answers
Final report quality and documentation (3)	this report, plus README, code walkthrough and handoff guide in the repository
Communication and presentation (2)	slide deck (submissions/) with agenda, timing and live demo
Viva / project defence (3)	every design choice is justified in Ch. 3–4; limitations and future scope in Ch. 7–8

Table 18: *Where each ESE rubric component is addressed*

Appendix B. Class Balance per Domain

Domain	Human	AI
hc3: reddit_eli5	4,623	4,623
hc3: finance	878	878
hc3: medicine	273	273
hc3: wiki_csai	191	191
hc3: open_qa	33	33
hape/acad	592	592
hape/blog	736	736
hape/fic	673	673
hape/news	637	637
hape/spok	830	830
hape/tvm	530	530
m4gt/arxiv	1,553	1,553
m4gt/reddit	1,389	1,389
m4gt/wikihow	1,385	1,385
m4gt/wikipedia	1,327	1,327
m4gt/peerread	196	196
m4gt/outfox	146	146
Total (17 domains)	**15,992**	**15,992**

Table 19: *Number of human and AI texts per domain after cleaning and balancing*